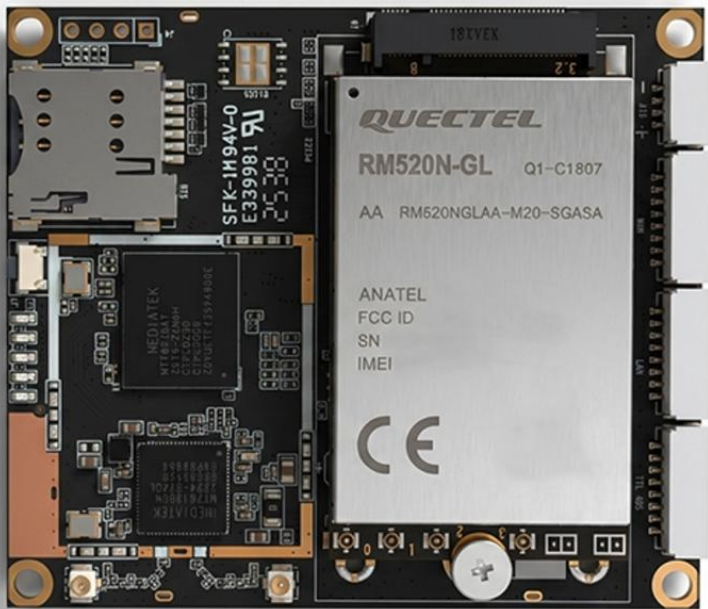


D228 Industrial 5G Router Module

Public Datasheet | Rev. 1.0

D228 is an industrial-grade LTE/NR gateway module for OEM equipment, industrial networking, remote device access, serial data applications, and compact embedded router deployments. In a board-level PCBA form factor it integrates 3G/4G/5G cellular access, dual Gigabit Ethernet, 5.8 GHz Wi-Fi, three serial ports (TTL/RS232/RS485), configurable IO, and remote management support.



Product positioning	Description
Product Type	Industrial LTE/NR router module with expansion interfaces
Target Customers	OEM manufacturers, system integrators, industrial IoT teams, and mobile equipment makers
Typical Projects	Remote monitoring, cellular failover, video backhaul, private-network access, serial (RS485/RS232/TTL) and I2C data acquisition, IO control, GNSS positioning
Supply Model	Evaluation samples and project configurations available by request

1. Key Features

- Board-level industrial LTE/NR router module for OEM integration and field-device connectivity.
- Cellular path based on the selected 5G/LTE module and regional project configuration, supporting 3G/4G/5G.
- Two Gigabit Ethernet interfaces; WAN/LAN roles depend on operating mode.
- Three serial interfaces (TTL, RS232, RS485) supporting transparent transmission, Modbus, MQTT, command mode, GPS/BeiDou data, HTTP POST, or custom protocol conversion.
- 5.8 GHz 802.11a/n/ac Wi-Fi for AP, client, bridge, and repeater deployments.
- Two configurable 3.3 V IO channels; I2C use supported by project configuration.
- 20Pin / 50Pin board-to-board expansion for Ethernet, serial, SIM, PCIe, USB 2.0, TF, reset, LED, IO, and power.
- Remote management platform support for status visibility, Web UI access, terminal access, tunneling, site-to-site networking, OTA workflows, and project integration.

2. Typical Applications

Use case	How D228 is used
Industrial Router Module	Embedded in customer equipment as a compact LTE/NR router, Ethernet gateway, Wi-Fi access point, and remote access node
Serial And IO Gateway	Connects PLCs, sensors, meters, controllers, and field devices to MQTT, HTTP, TCP/UDP, Modbus, or customer platforms
Mobile And Field Equipment	Provides cellular access, Wi-Fi connectivity, VPN/private networking, and remote diagnostics for distributed field assets
OEM Remote Operations	Supports remote login, Web UI access, terminal sessions, tunneling, status reporting, OTA upgrade, and platform-based maintenance

3. Core Specifications

Item	Public specification
Cellular Module	Standard reference configuration may use Quectel RM520N-GL 5G module; regional cellular

	module options may vary by project
Cellular Standards	3G / 4G / 5G; final support depends on selected module, firmware, carrier requirements, and target region
Reference Bands	Reference configuration: NR NSA n41/n78/n79; NR SA n1/n28*/n41/n77/n78/n79; FDD LTE B1/B2/B3/B5/B7/B8/B20/B28; TDD LTE B34/B38/B39/B40/B41; WCDMA B1/B2/B5/B8. Final band support depends on selected module and region
Ethernet	On-board 2 x 1000 Mbps Gigabit Ethernet; WAN/LAN roles depend on operating mode
Wi-Fi	5.8 GHz 802.11a/n/ac 2T2R, up to 866 Mbps
Serial	Three serial ports: TTL, RS232, and RS485 via UART External; final pinout and multiplexing follow the project interface definition
IO	2 x configurable 3.3 V IO channels; I2C use supported by project configuration
SIM	Micro SIM slot; notch faces outward, contacts toward the PCB. Do not hot-plug SIM while powered
Antennas	2 x Wi-Fi IPEX-1; 4 x LTE/5G IPEX-4; optional GPS IPEX-1 depending on model/project configuration
Power Input	Two 4Pin 1.25 mm power connector positions (horizontal/vertical); use either one. Input 7-15 V, recommended 12 V / 2 A. 5 V input support requires project confirmation
Typical Power	Below 5 W under full-load reference conditions; reserve at least 5 W supply budget for cellular peak load
Operating Temperature	Supports -40 °C cold start and long-term operation up to +70 °C
Board Outline	Approx. 72.0 mm × 61.7 mm × 15.0 mm; mounting ear hole spacing approx. 67.1 mm × 56.7 mm with 4 locating/mounting holes. Confirm against mechanical drawing before enclosure design
Expansion Connectors	Rear dual-row B2B, 1.0 mm pitch, 20-pin + 50-pin (40 + 100 contacts); expands Ethernet, serial, SIM, PCIe, USB 2.0, TF, reset, LEDs, I/O, and power

4. Product Views and Interface Overview

This section provides front/rear product views, interface layout, and mechanical dimension references for product identification, connector location checks, and early enclosure integration review. Complete pin definitions, electrical thresholds, grounding/shielding, protection design, connector height, enclosure clearance, and thermal conditions should be confirmed during the project engineering phase.

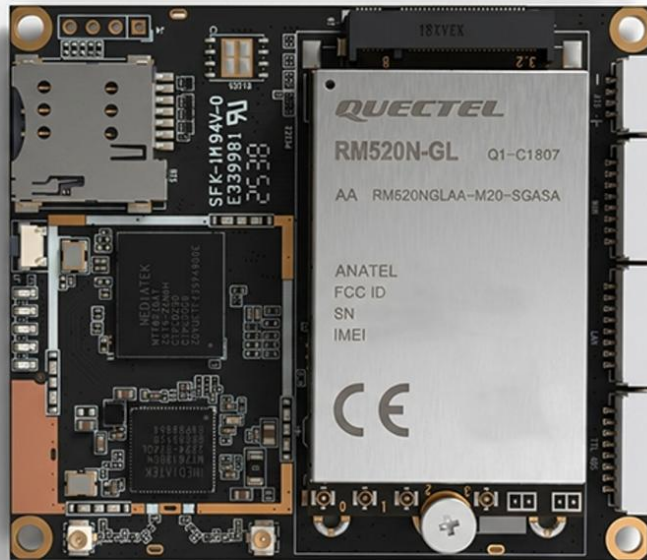


Figure 1. Front product view

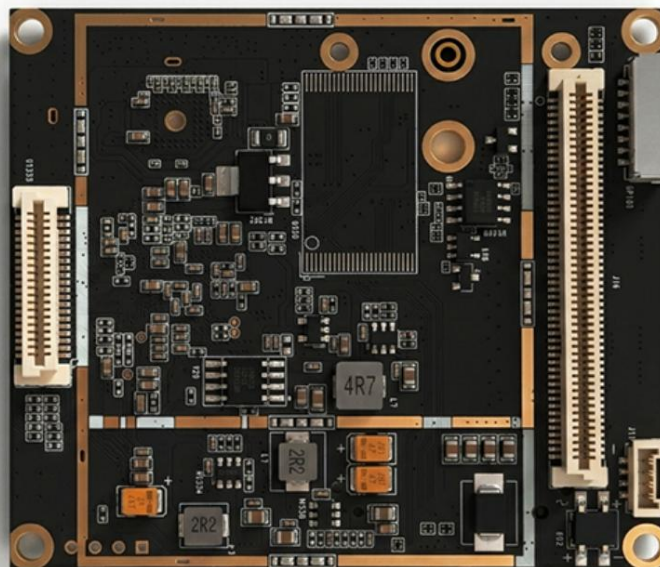


Figure 2. Rear product view

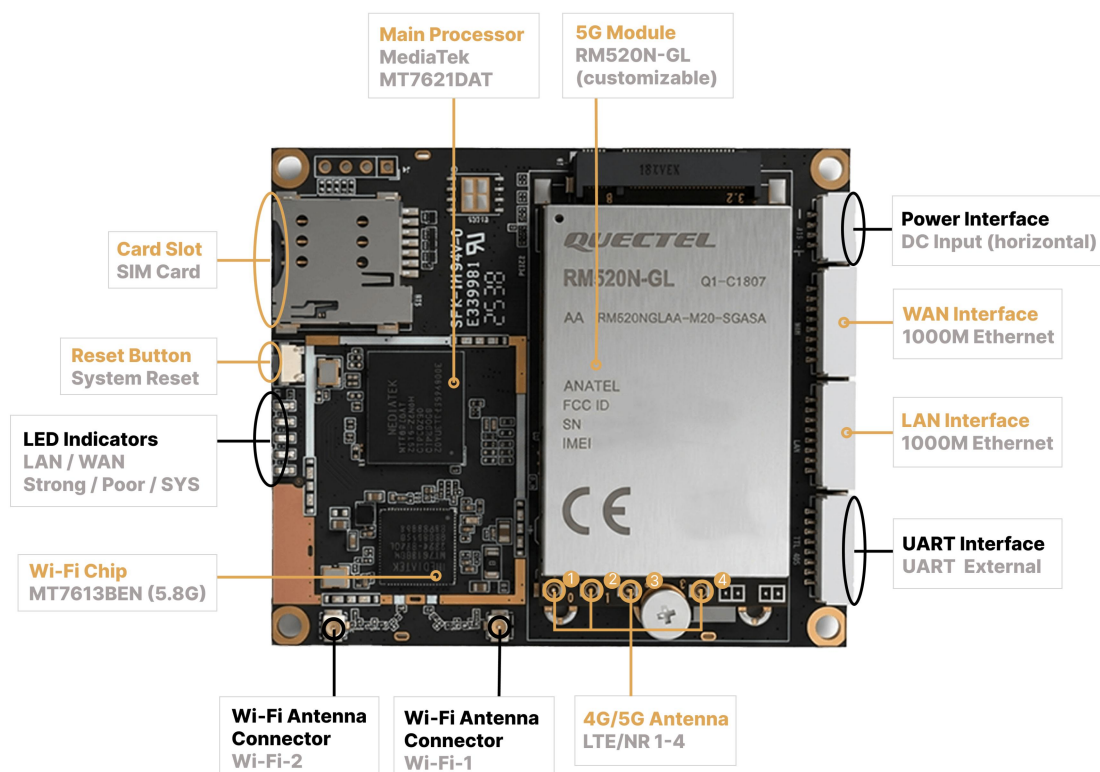


Figure 3. Front-side interface and major component layout

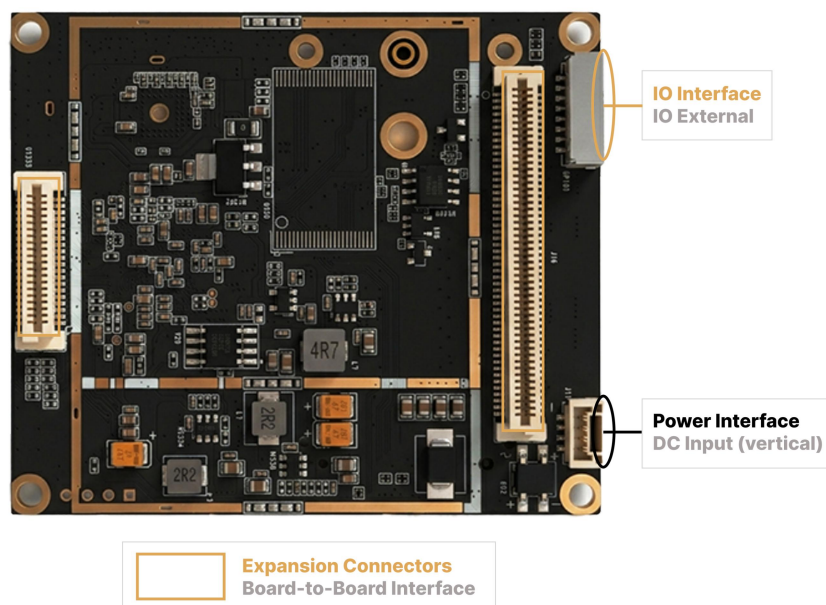


Figure 4. Rear-side interface and functional area layout

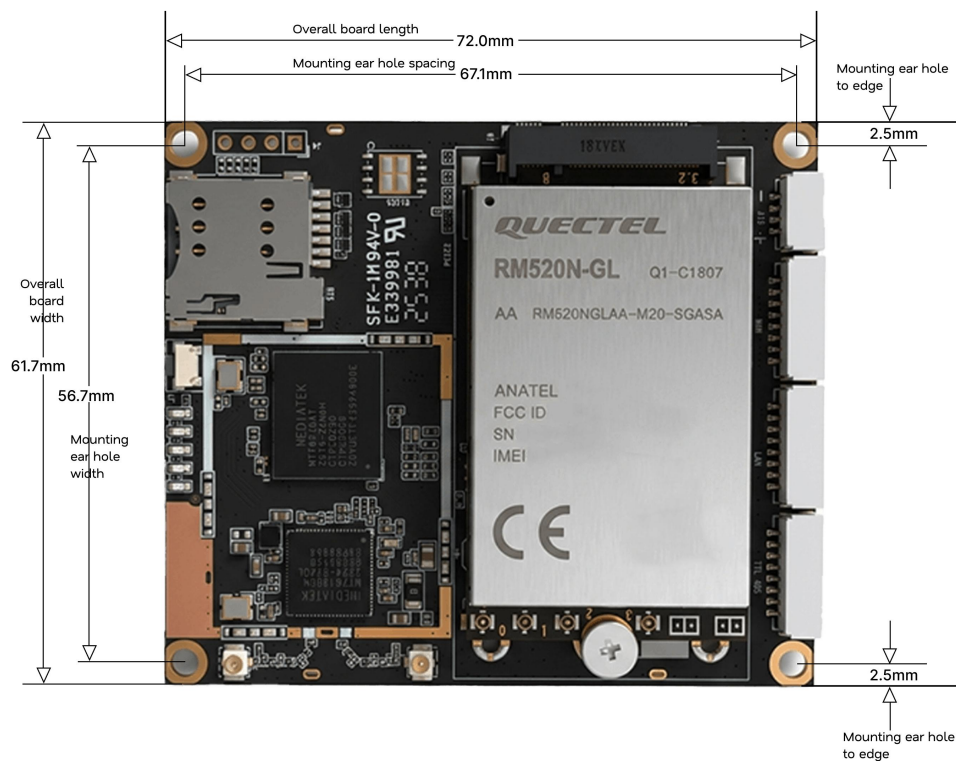


Figure 5. Front-side mechanical dimensions and mounting reference

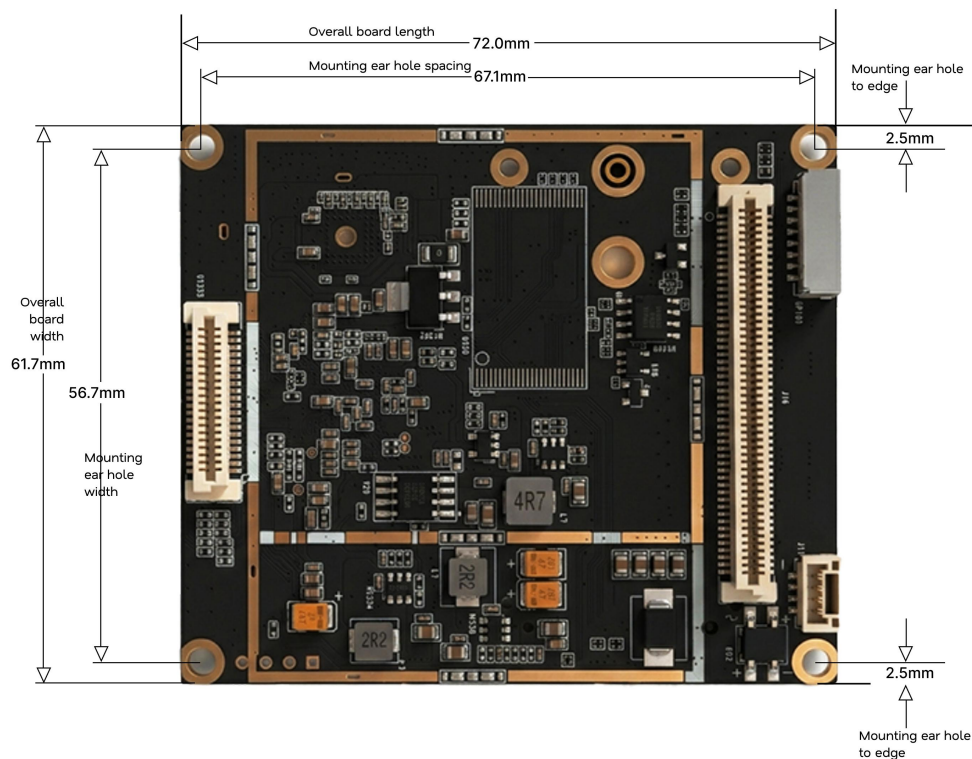


Figure 6. Rear-side mechanical dimensions and mounting reference

Interface group	Public summary
Power	Horizontal and vertical 4-Pin 1.25 mm DC input positions; use the defined input position for the project design. Supports 7-15 V
Ethernet	WAN and LAN 1000 Mbps Gigabit interfaces; additional Ethernet signals may be accessed through the expansion connector with proper transformer design
Serial / IO	UART External provides TTL, RS232, RS485, and configurable IO lines for serial data, field-device connection, industrial control, and project-defined behavior
SIM / Reset / Indicators	Micro SIM, reset, LAN/WAN/Strong/Poor/SYS indicators, and related signals support field install and maintenance workflows
Expansion Connectors	20Pin / 50Pin board-to-board connector can provide extended Ethernet, serial, SIM, PCIe, USB 2.0, TF, reset, LED, IO, and power access. Detailed pin definition is provided during project engineering review

Engineering package available by project

During project evaluation, detailed pin definitions, electrical thresholds, connector heights, component heights, DXF/STEP reference files, protection notes, and enclosure integration support can be provided on request.

5. Model Options

Model	Description
D228 Standard	Integrated 4G/5G LTE/NR cellular module with dual Gigabit Ethernet, three serial ports, IO, Wi-Fi, and remote management
D228-4G	Integrated 4G LTE cellular module for projects that require 4G LTE access only
D228-GPS	Integrated 4G LTE cellular module with GPS positioning for projects that need location data collection and reporting
D228 Core	M.2 socket for customer-selected 5G NR module; module compatibility, antenna routing, firmware support, thermal behavior, and certification scope require project review
D228-4G Core	MiniPCIe socket for customer-selected 4G LTE module; intended for project-specified or already-certified module adoption

D228-NAND	Integrated 4G/5G LTE/NR cellular module with NAND Flash for larger storage, software package deployment, or custom development environments
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6. Software and Remote Management

- Talonbox SkinOS embedded platform supports modular development, software package deployment, SDK access, and project-level customization.
- Supports local TCP control, local HTTP status query, LAN discovery, SMS management, remote HTTP/MQTT reporting, and platform-based device management.
- Supports remote Web UI access, terminal access, network tunneling, OTA upgrade, configuration management, log viewing, platform API, and maintenance workflows.
- Depending on firmware and project configuration, supports cellular, wired WAN, Wi-Fi client/repeater, bridge/forwarding, and hybrid multi-WAN operation.
- Serial protocols include transparent transmission, Modbus RTU to Modbus TCP, Modbus data acquisition, MQTT, command mode, GPS/BeiDou data, HTTP POST, and custom protocol conversion.
- Where supported by configuration, firewall, NAT/DMZ, port proxy, static routing, policy routing, SNMP, DDNS, UPnP, PPTP, L2TP, IPsec, GRE, OpenVPN, WireGuard, NAT traversal, and high-availability features are available.

7. Ordering and Project Notes

Item	Public notes
Evaluation Samples	Available by project request; provide target region, carrier, antenna plan, interface needs, and expected application
OEM Configuration	Firmware defaults, remote platform settings, serial behavior, IO definition, protocol conversion, branding, and interface options can be reviewed by project
Lead Time	Eligible standard configurations may start from 7 days. Final timing depends on quantity, cellular module option, certification, connector/interface configuration, and customization scope
MOQ	Sample and production MOQ confirmed per project

Supporting Files	Mechanical drawings, interface layout, connector references, and engineering package materials can be provided during project evaluation
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8. Certification and Configuration Notes

- Certification status, cellular module options, antenna configuration, supported bands, product labeling, reference images, carrier acceptance, access permissions, encryption policy, platform settings, and lead time may vary by target market and project configuration.
- Module-level certification marks shown in reference images apply to the cellular module only and do not automatically indicate certification of the D228 board-level product or the end system.
- The customer end product—including enclosure, antenna design, power input, connector/interface design, serial/IO routing, mounting method, labeling, and regional deployment requirements—may require additional engineering review or certification.

9. Contact and Project Evaluation

For a datasheet package, evaluation samples, OEM integration review, mechanical files, connector references, or remote management support, send your project requirements to Talonbox and include target region, carrier requirements, enclosure plan, and required interfaces.

Recommended information to include

Target region, carrier requirements, expected quantity, enclosure plan, required interfaces, serial/IO requirements, and whether the project needs a standard D228 configuration or OEM integration review.

- Email: sales@talonbox.com
- Website: <https://www.talonbox.com>
- Inquiry form: <https://www.talonbox.com/#contact>

Final configuration, lead time, certification scope, and engineering file availability are subject to project confirmation.